

A

Preparation and query path

static extraction, no model call

parsers rank files and resolve imports;
Tree-sitter and Joern supply structure

queryable evidence store

computations and result anchors; conditions;
definitions; call boundaries with providers

endpoints, beside shell tools

science_find / inspect / note

B

Recorded payloads

boundary record, task 103, shell.py:1005

```
{"callee": "radians", "code": "np.radians(thetad)",  
  "boundary": {"kind": "external",  
               "provider": "numpy",  
               "assume": "correct_interface",  
               "repair_scope": "caller_file"}}
```

search response (keys verbatim)

```
{"status": "ok", "matches": [{"target": ...,  
  "excerpt": ..., "kind": "scientific_node"}],  
  "scope": "Lexical discovery, not a verdict."}
```

same call in the baseline arm

```
grep -rn "radians" .      # provider inferred by hand
```